

Abstract No.

Glioblastoma with diverse oncogenic drivers share translationally relevant vascular elements.

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The enormous inter- and intra-individual heterogeneity of glioblastoma (GBM), both on the level of tumor genetics and cellular composition, represents a major caveat for successful clinical management. However, identification of conserved neurodevelopmental pattern within the diverse cellular and genetic landscape of GBM raised hopes for the identification of therapeutic targets with broader impact for this entity. We hypothesized that glioma stem cell (GSC) differentiation may represent another, common evolutionary element in gliomagenesis. We generated a transgenic, lineage-tracing mouse model for GBM in which GSCs are readily identifiable by a fluorescent reporter. Furthermore, the model permits investigating diverse GBM driver mutations generating tumor subsets that differ e.g. in overall survival. Histopathological inspection revealed that the intratumoral vascular pattern also varied among genetically distinct GBM. However, lineage identification of intratumoral vascular mural cells (VMCs) demonstrated that genetically distinct GBM commonly harbored a large population of tumor-derived VMCs (td-VMCs). Orthotopic transplantation of GSCs generated GBM with td-VMCs, but GBM cell-lines did not. Conditional knockout of platelet derived growth factor receptor-B (PDGFRB) in the GBM primordium prevented td-VMC generation, suggesting that td-VMC formation recapitulated a neurodevelopmental process. Lineage ablation of td-VMCs resulted in vascular normalization and increased permeability, while targeting host-derived VMCs had adverse effects. Altogether, we show that td-VMCs have broad therapeutic relevance for genetically diverse GBM and that adjuvant treatments for td-VMCs can overcome therapy resistance imposed by the blood-tumor barrier.



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